

CITS4403 COMPUTATIONAL MODELLING · RESEARCH PROJECT

Chokepoint Disruption in the Global Oil Network

A network cascade model of the 2026 Strait of Hormuz closure: where does a local blockage become a global shortage?

Raed · [demo date]

THE SYSTEM

One strait carries a fifth of the world's oil

20.9

mb/d through Hormuz in 2023
(EIA)

~20%

of world petroleum consumption

4.9

mb/d still flowing in Q2 2026
after the closure (EIA)

Bypass pipelines can carry only part of it, and several Gulf producers have no bypass at all. Brent rose from about **\$69** to about **\$105**.

RESEARCH QUESTION

As more of Hormuz is blocked, is there a critical closure level at which a local shortfall becomes systemic production failure, and which intervention raises that threshold most?

Interventions compared: bypass pipeline capacity, strategic reserve release and demand restraint. Also tested: what a second closure, at Bab el-Mandeb, adds.

MODELLING CHOICE

Why a capacitated flow network

Framework	Fit for this question
Cellular automaton	Needs a spatial lattice; oil trade is not spatially local
Agent-based model	Suits many heterogeneous actors; here physical capacity binds
System dynamics	Aggregate stocks, no network; the question is about structure
Capacitated flow network	Chosen: oil moves through capacity-limited conduits

FROM THE LECTURES

Graph $G = (V, E)$: who connects to whom. ER, WS and BA generators.



WHAT THIS MODEL ADDS

Edge capacity, edge direction, and node state that changes each day.

THE MODEL

Three coupled layers, run every simulated day



The layers are different networks: Layer 1's nodes are places, Layer 3's are industries. They connect through one number, the fraction of needed oil each region receives.

THE MODEL

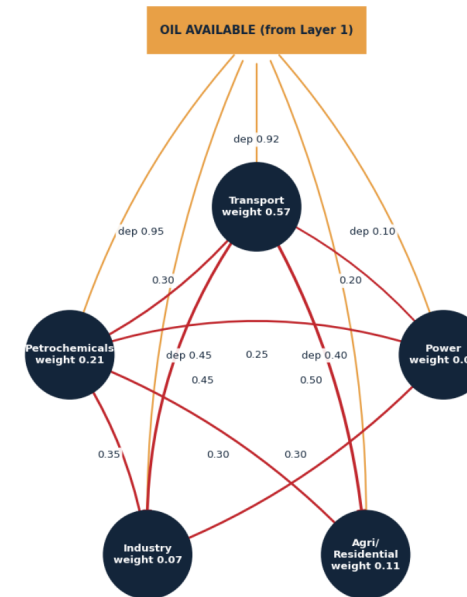
Layers 2 and 3 are networks too

Layer 2 - the market: reserves, then the price



Circles + lines = `clear_market()` in `oil_network_model.py`: regions' demand (elasticity e) $\rightarrow$ ONE price.
Squares = strategic reserves, applied BEFORE it in `spr_depletion.py`: each covers ONLY its own region's shortfall.
Demand cut by price is NOT sent back to Layer 1.

Layer 3 - the production cascade, one region (`run_production_cascade`)



Amber arrows: how oil-dependent each sector is. Red arrows: supplier sector $\rightarrow$ user sector, labelled with how much the user needs it. Each sector: $\text{output} = \min(\text{direct, upstream}) \times \text{gate}$, gate shuts a sector below 45% of inputs. OUT: $\text{systemic loss} = \sum(\text{weight} \times \text{lost output})$

LAYER 1 DATA

Ten regions, each producing and consuming

Region	Produces	Consumes	Net
North America	31.1	24.6	+6.5
Middle East	31.0	9.5	+21.5
CIS	13.8	4.8	+9.0
Latin America	8.4	6.5	+1.9
Africa	7.6	4.4	+3.2
China	5.3	16.4	-11.1
Europe	4.0	14.5	-10.5
Other Asia	3.1	10.0	-6.9
India	0.9	5.6	-4.7
Japan/Korea	0.1	5.7	-5.6
World	105.3	102.0	

Total liquids

Crude, NGLs, biofuels and refinery gain on both sides, in mb/d

~20%

Hormuz's share of the modelled system, matching reality

Domestic first

Each region uses its own production before anything is traded

HOW THE CALCULATION TRAVELS

One simulated day, step by step

- 0 **Apply the closure.** Hormuz capacity $\times$ open fraction. This is the only input that changes.
- 1 **Domestic first.** Each region consumes its own production; that oil can never be bid away.
- 2 **Split the surplus.** Exports divide into a contracted share and a spot share.
- 3 **Stage 1, contracted.** Fill the fastest routes first, limited by pipelines, canals and the closure.
- 4 **Stage 2, spot.** Remaining oil goes to the highest netback: scarcity bid minus freight.
- 5 **Score the day.** Unserved = demand – delivered, which drives price and the production cascade.

Four formulas carry the model

Route capacity

$$\text{headroom} = \min(\text{cap} - \text{used})$$

A route is limited by its tightest pipeline or canal.

Bidding

$$\text{premium} = 1 + 4 \cdot (\text{short}/D)^2$$

$$\text{netback} = \text{premium} - 0.01 \cdot \text{days}$$

Desperation decides; distance barely matters.

Price

$$\text{demand} = D \cdot \text{price}^e$$

$$\text{solve } \sum D \cdot \text{price}^e = \text{supply}$$

Inelastic demand ($e \approx -0.1$) makes price steep.

Production

$$\text{out} = \text{gate} \cdot \min(\text{own}, \text{upstream})$$

$$\text{loss} = \sum \text{share} \cdot (1 - \text{out})$$

A factory with fuel still stops without trucks.

THE CODE

Nine files: data, model, analysis, tests

File	What it does	Key function
oil_model_data.py	Every number, labelled data or assumption	constants only
oil_network_model.py	Layers 1–2: routing, allocation, market	allocate_flows, step
production_cascade.py	Layer 3 cascade, threshold, interventions	compare_interventions
spr_depletion.py	Regional strategic reserves	simulate_with_reserve
price_dynamics.py	Price path over time	simulate_price
closure_sensitivity.py	Hormuz × Bab el-Mandeb closure grid	steady_state, grid
sensitivity.py	Parameter sweeps, path dependence	path_dependence_test
topology_comparison.py	Network metrics vs ER, WS, BA	comparison_networks
test_model.py	23 correctness and boundary checks	test_conservation

HOW A RUN TRAVELS

From notebook cell to result

- | | | |
|---|---|----------------------------|
| 1 | <code>m = OilNetworkModel()</code> | state set to normal |
| 2 | <code>for day in range(150): m.step(f)</code> | one simulated day |
| 3 | <code>allocate_flows(f)</code> | domestic, stage 1, stage 2 |
| 4 | <code>update refineries, hoarding</code> | state carried forward |
| 5 | <code>clear_market(supply)</code> | price from shortfall |
| 6 | <code>regional_cascade(f)</code> | shortfall to sectors |
| 7 | <code>critical_blockade_intensity()</code> | bisection: 53% |

Four nested loops

Scenarios. 21 closure levels, or the 11×11 grid. A fresh model each.

Days. 150 steps, so state can evolve.

Routes. 37 per day, one greedy pass.

Fixed point. Sector outputs, until they stop changing.

IS IT RIGHT?

Only quantities the model is not given count as tests

NOT A TEST

4.9 vs 4.9 mb/d

Hormuz flow. The closure level was derived from the observed 4.9, so Hormuz simply fills to that ceiling.

THE GENUINE TEST

\$94–\$143

Brent at the Q2 2026 point, with and without reserves. Observed: about \$105, inside the bracket.

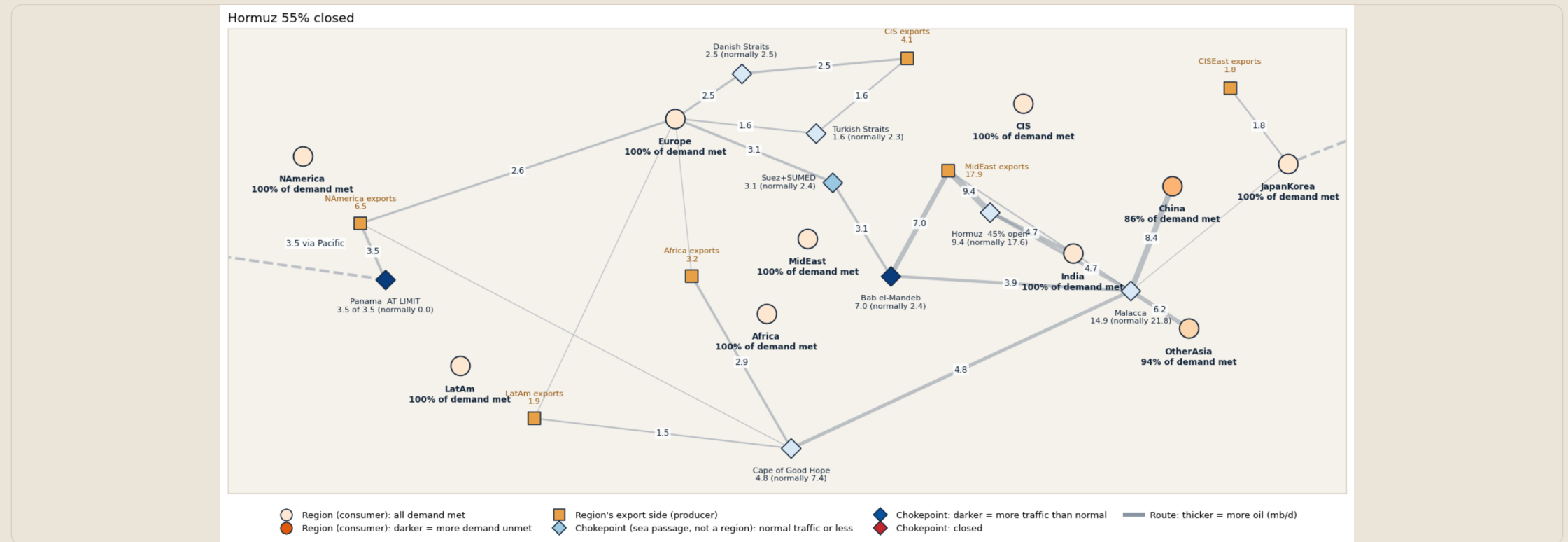
Corridor	Model	Real crude	Verdict
Panama	0.0	~0.1–0.2	Correct: crude tankers cannot fit
Suez	2.4	~3.7	Under by about a third
Bab el-Mandeb	2.4	~6.2	Under by more than half

Below the threshold, the network absorbs the loss



THE NETWORK AT 55% CLOSURE

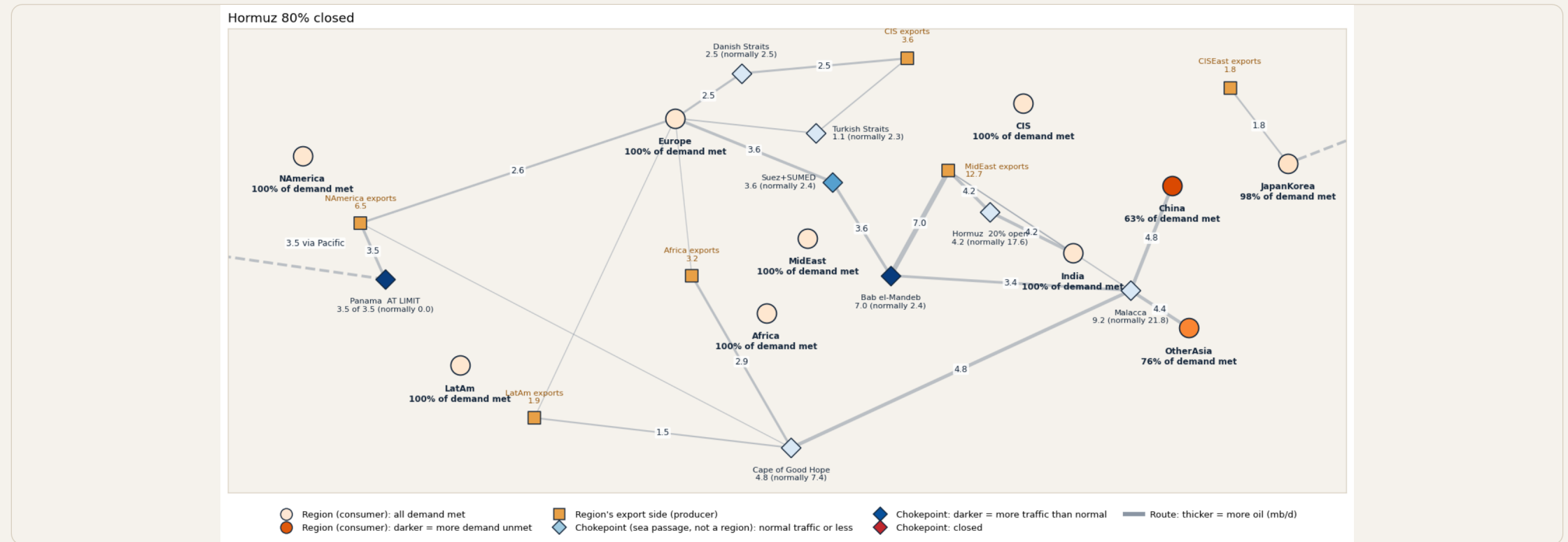
Just past the threshold, Asia starts to run short



China is 86% met and Other Asia 94%. Bab el-Mandeb and Panama now carry far more than normal.

THE NETWORK AT 80% CLOSURE

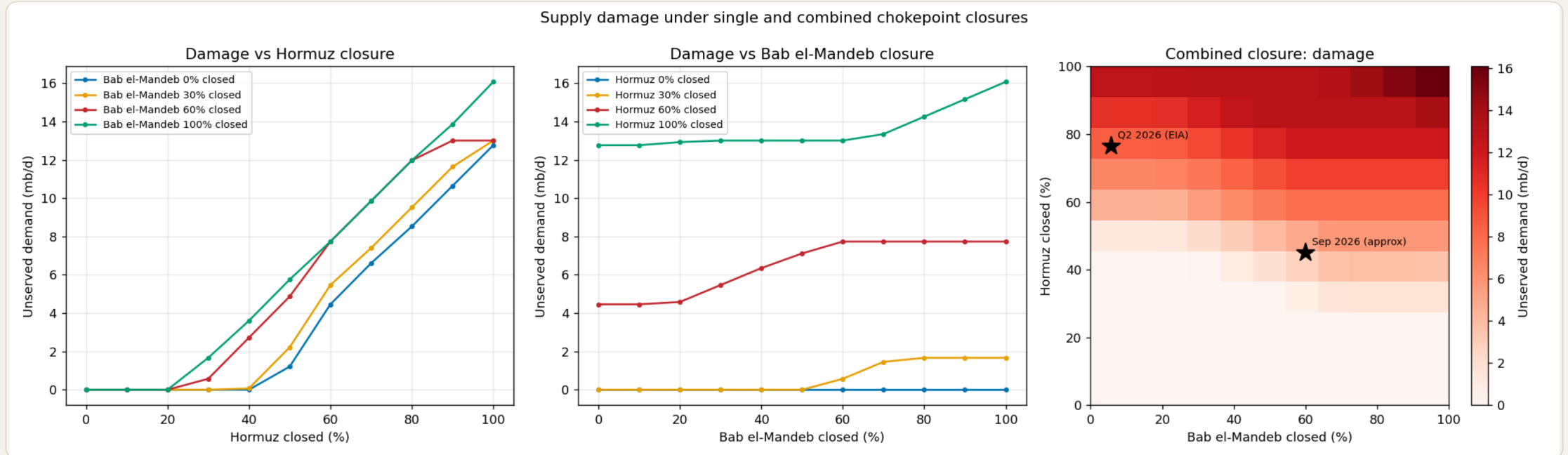
Deep closure: every escape route is full



China falls to 63% and Other Asia to 76%. The pipelines and Panama are at their limits.

CLOSURE SENSITIVITY

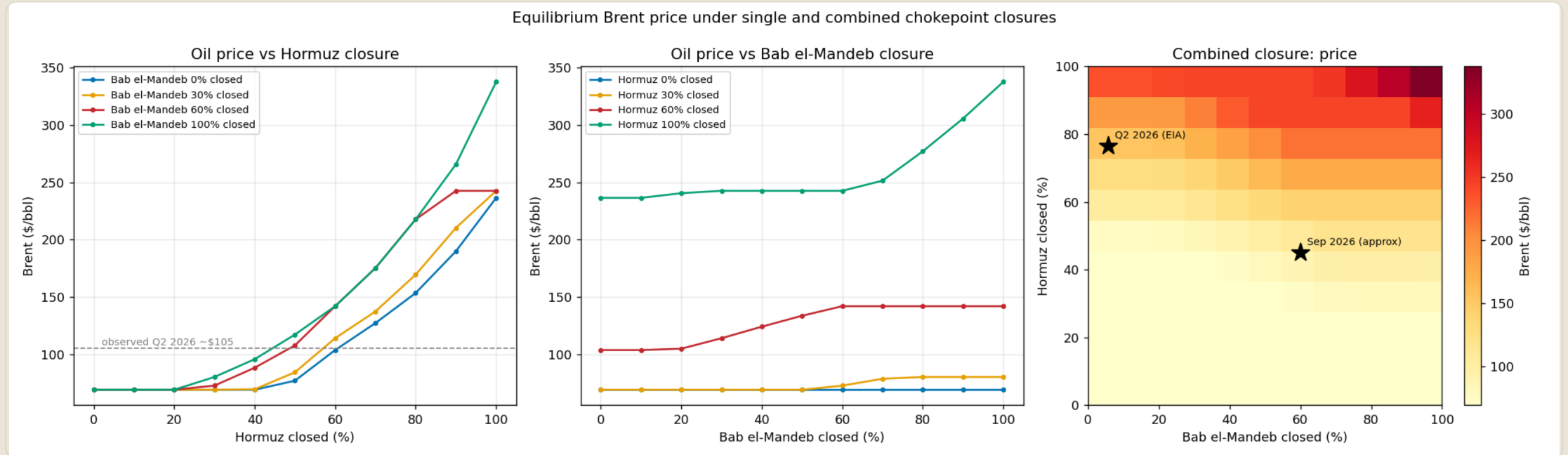
Hormuz drives the damage; Bab el-Mandeb amplifies it



Little damage until about half of Hormuz closes. At 70% Hormuz closure, also closing Bab el-Mandeb raises unserved demand from **6.6 to 9.9 mb/d (+49%)**.

CLOSURE SENSITIVITY

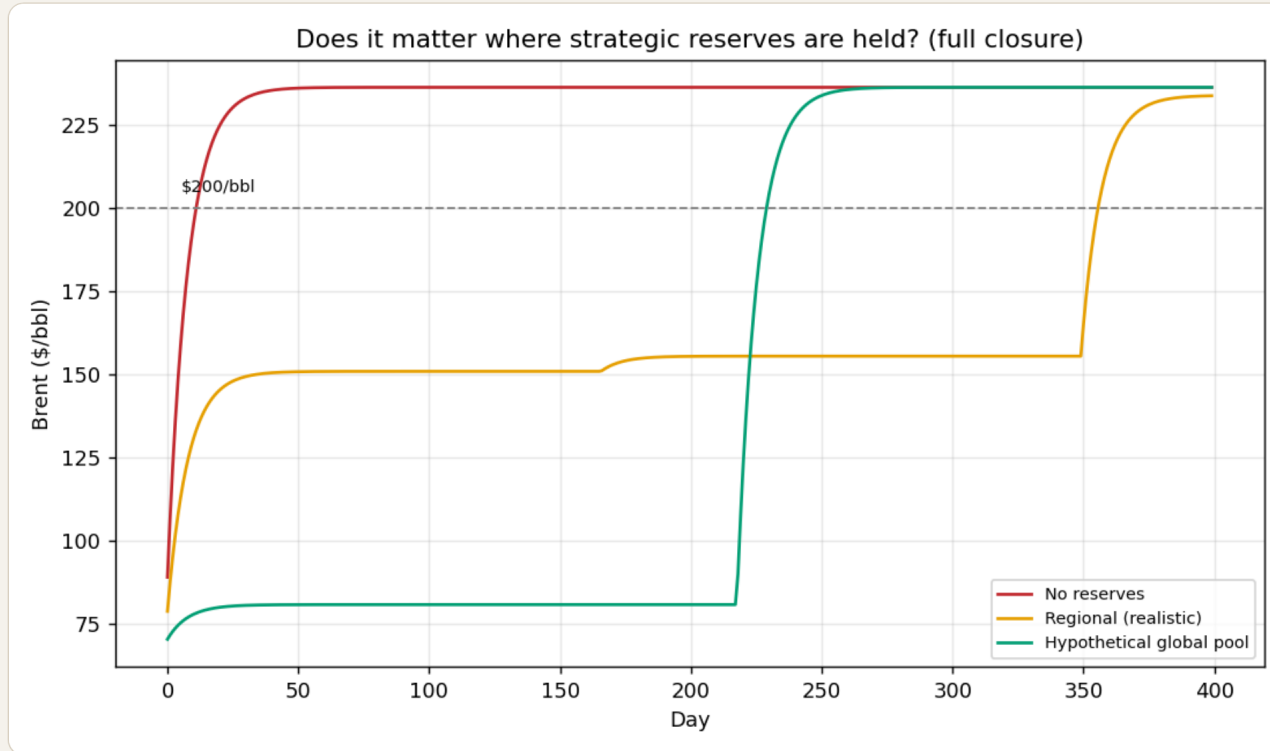
Price rises steeply because demand barely responds



Brent stays near \$69 until about 30% closure, then curves upward: each extra barrel of shortfall costs more than the last. These prices assume **no** strategic reserves.

INTERVENTION

Strategic reserves sit in the wrong place



BRENT AT DAY 100, FULL CLOSURE

\$236

No reserves

\$151

Realistic: each region uses only its own

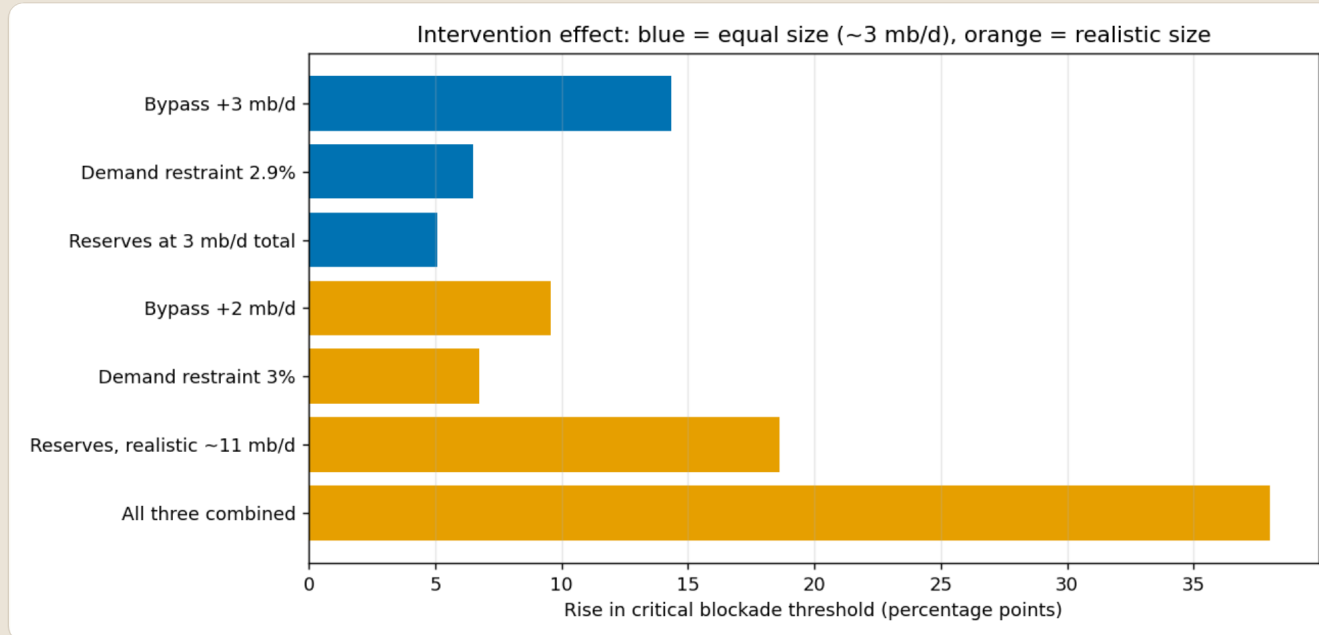
\$81

If all ~2,400 Mb could be pooled

~500 Mb (US, Europe, India) is never drawn: those regions are not short.

ANSWERING THE QUESTION

Which intervention raises the threshold most?



PER BARREL, ~3 MB/D EACH

Bypass capacity +14 pts

Demand restraint +7 pts

Reserves +5 pts

REALISTIC SIZE

Reserves, ~11 mb/d +19 pts

Bypass, +2 mb/d +10 pts

Demand restraint, 3% +7 pts

Reserves win in practice only while they last: at 71% closure China's reserve runs out on **day 350** and the system turns systemic the next day. Bypass and restraint do not expire.

RESULTS

What the model shows

53%

Critical blockade

Hormuz closure where the worst region first loses 10% of output. An et al. report 32–46% on a different definition.

+49%

Correlated failure

Damage added by closing Bab el-Mandeb at 70% Hormuz closure, because the main bypass exits through it.

5.6% vs 12–27%

More robust than random

Capacity lost when the most central chokepoint is removed: real network vs ER, WS, BA and rewired graphs.

+31%

Volatility matters

Extra damage from on-off disruption versus a smooth ramp of equal total exposure. Order alone: none.

MODEL REVIEW

Errors found and corrected along the way

Fixed

- Crude-only production compared with total-liquids consumption
- European demand entered as Germany $\times 4$ (8.2 instead of 14.5 mb/d)
- Russian Pacific crude routed through Malacca, and a Gulf-to-Asia route via the Cape
- 10 mb/d pushed through a 5 mb/d bypass pipeline
- No bypass routes to Japan/Korea or Other Asia, which made bypass look useless
- Hormuz flow set from observation, then called validation
- Price counted demand destruction twice, understating it

RETRACTED FINDING

"North America is hurt by contagion"

An artefact of the crude-versus-liquids error. On a consistent basis North America produces 31.1 and consumes 24.6 mb/d: a net exporter, not exposed.

[Acknowledgement of AI assistance, as required by UWA policy for this unit]

LIMITATIONS

What the model cannot yet claim

Limitations

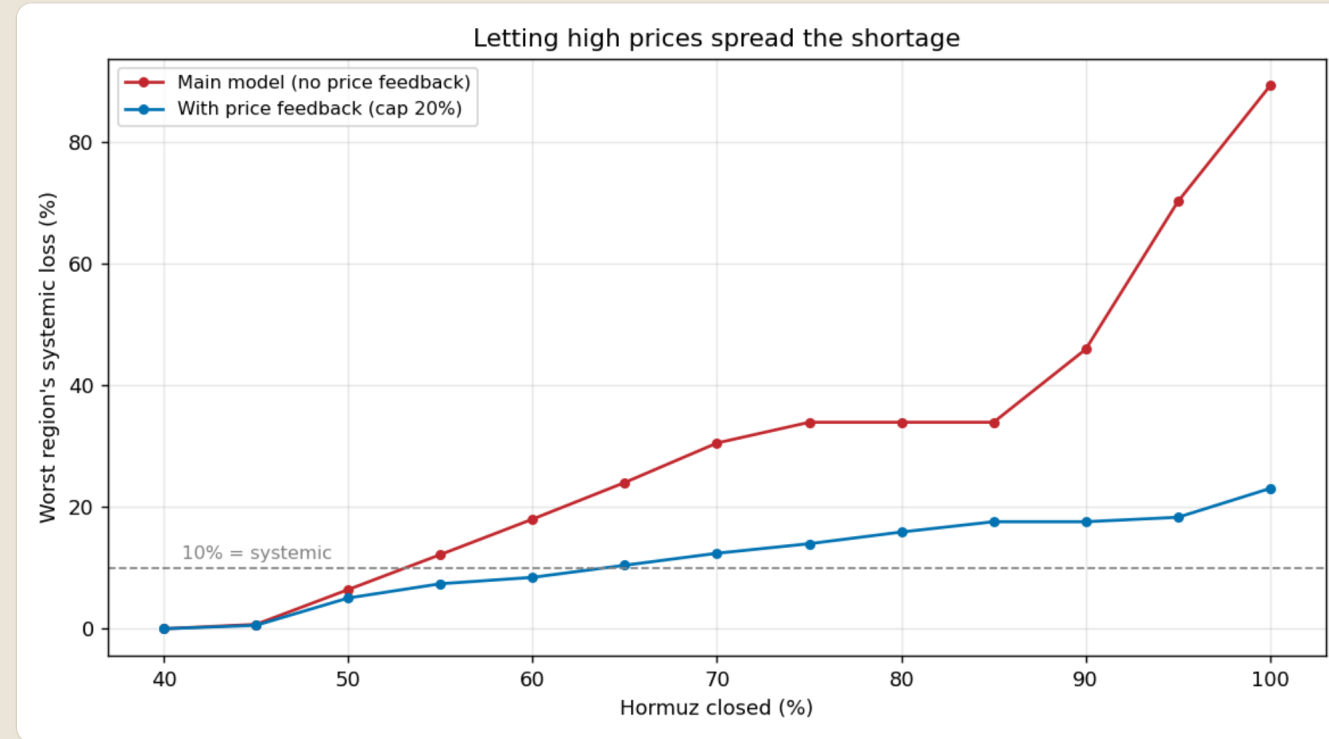
- Greedy routing, no price feedback: oil freed by high prices never reaches Asia
- Routes are hand-built: a review found missing bypass legs, and others may remain
- No refined-product trade, so Suez, Bab el-Mandeb and Panama are under-represented
- Straits have no limit, only a normal flow; canal limits are assumptions
- Hoarding and friction disabled; demand restraint treated as costless
- Refinery-complexity uplift is a proxy: without it the threshold is 51%

Next steps

- Equilibrium routing: spread trade as freight on crowded routes rises
- Long-term contracts and terminal-level origins
- Refined products as a second commodity
- Hoarding as inventory that moves price, not allocation

EXTENSION: FIXING THE BIGGEST LIMITATION

Letting high prices spread the shortage



53% → 64%

Threshold, main model → with price feedback

89% → 23%

Worst region's loss at full closure

Everyone cuts

India and Africa to 80% of their oil, Europe 89%, China 81%

THE ANSWER

Systemic failure begins near 53% closure. Per barrel, bypass capacity raises that threshold most; in practice reserves do, but only until they run out.

Below the threshold, pipelines and other suppliers absorb the loss; above it, shortage concentrates in Asia and price climbs steeply. The bypass is only as safe as Bab el-Mandeb.

Questions